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Lecture 6: Solvency Capital Requirements – Backtest, IFRS and alternative Concepts

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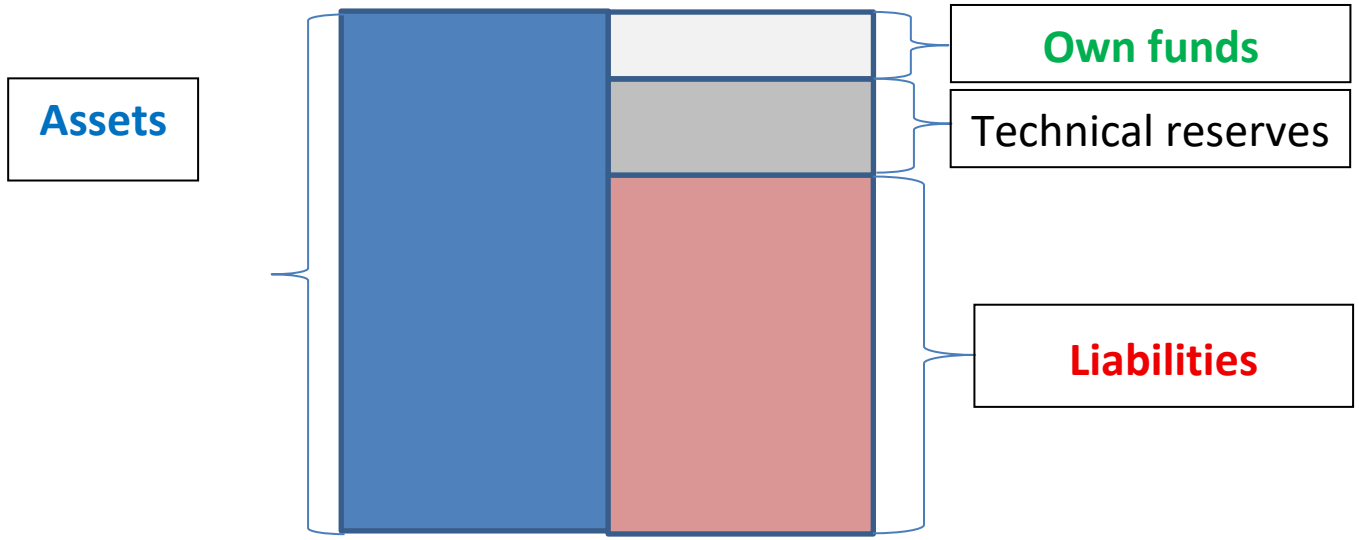
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Mathematics for Solvency Capital Requirements:

- Legal and conceptual background
- Backtesting of Solvency II
  - Data analysis
  - Data analysis II
  - Top-down vs. bottom-up
- Solvency balance sheet vs. IFRS balance sheet
- Portfolio optimization under Value-at-Risk constraints

Solvency II – A simplified balance sheet

To cover its liabilities, every insurer has to put aside so much capital at the start of the year such that the insurer is **solvent by the end of the year** with **a probability of at least** 99,5%.



Main question: How big have to be the own funds to satisfy the above probability requirement? How to compute this?

Solvency II – Concept, legal background and backtesting

- Solvency II introduced a **solvency balance sheet** and solvency requirements via the required solvency capital (**SCR**).
- From a mathematical point of view, it was like a financial mathematician's dream come true by introducing a **bottom-up approach** ("every part of a company is modelled") both in the **standard model** and in **internal models** where companys can use their own, however approved, model for their solvency calculations
- It states an exact variable that has to be presented (the **SCR**)
- … and on top of that there is even a precise formula how to combine the different SCR of the parts of the company to obtain the company/group SCR, the **Standard Formula**

So, what more could an actuary on the practical side and an actuarial mathematician in academia expect from new regulations?

Let us see …

Solvency II - Regulations for reliable Insurance Companies

European Council Directive 2009/138/EC

Article 122:

"Where practicable, insurance and reinsurance undertakings shall derive the solvency capital requirement directly from the **probability distribution forecast** generated by the **internal model** of those undertakings, using the Value-at-Risk measure set out in Article 101(3)."

- ⇨ One has to predict the **complete distribution function** generated by the internal model!
- ⇨ One has to simulate the **distribution of the random factors** and the resulting difference **„Liabilities - Assets"** of the insurer !!!

Still, everything sounds good!

Solvency II – The Concept of the SCR and its computation

Main conceptional idea: Define the own funds as saldo of the solvency balance sheet

(1) $$OF_t = A_t - L_t$$

where A_t is the market value of the assets and L_t the market value of the liabilities. The solvability of the company is then measured by the increment of the own funds

(2) $$\Delta OF_t = OF_{t+1} - OF_t$$

Given that the own funds can be expressed by the value of the portfolio π of the company at the different times

(3) $$OF_t = v(\pi_t)$$

with v a valuation operator, Equation (2) would then correctly be given as

(4) $$\Delta OF_t = v(\pi_{t+1}) - v(\pi_t)$$

Main problem of Equation (4): How to get/choose π_{t+1} ?

Solvency II – The Concept of the SCR and its computation – 2

Current practise: A purely static approach, no portfolio change

More precisely:

- Keep the asset and the liability positions fixed
- Project them one year into the future
- Determine the distributions of the asset and liability positions (better, the distribution of their differences)
- Infer the SCR as the 0,5%-quantile of the distribution function, i.e. determine SCR such that we have

(5) $$P_{t+1}(0 \leq \Delta v(\pi_t) + SCR) = 0{,}995$$

with P_{t+1} being the probability distribution of the (difference of the) future asset and liability positions.

This is quite a hard task and – given remaining time – we will possibly talk about that later ….

Solvency backtest – Data from a large German insurance group

Data and statistics based on/from K., Stahl (2023a), (2023b), K., Oppold, Schelling, Stahl (2026)

What is a backtest?

A backtest is an **empirical evaluation** of a hypothesis, a method or a model used/ assumed in the past. Using the history of corresponding data, one tries to judge if the hypothesis was justified or if the model/method did work.

In the Solvency case:

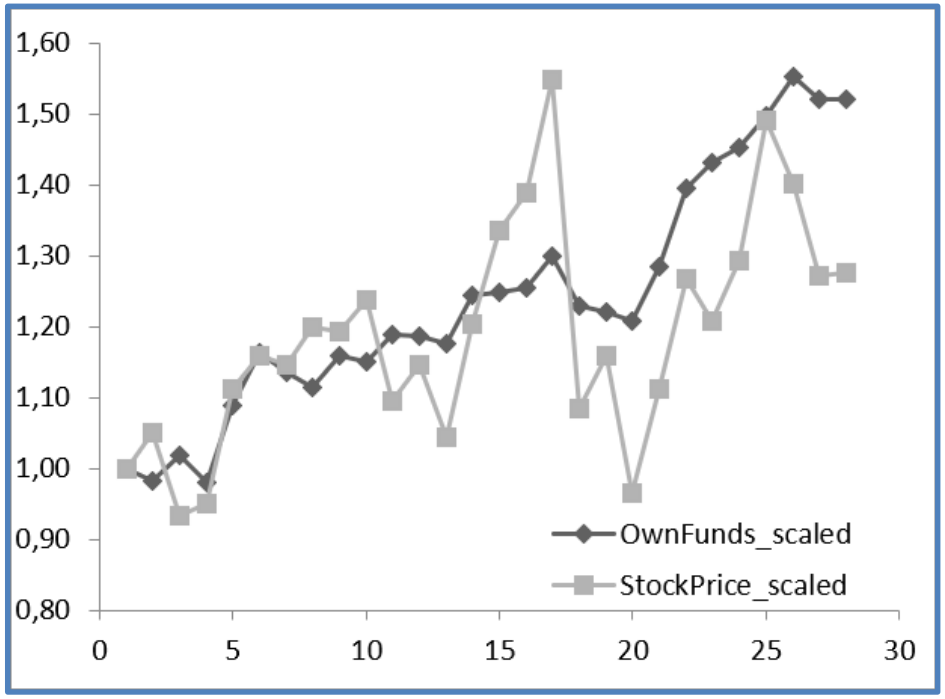
Use **all reported SCR values** of an insurance company or group and check **if they could have indeed been 99,5% quantiles** of the loss function where the time horizon was one year.

Problem: At the time of the backtest there typically had only been 28 reported SCR values (i.e. seven years with quarterly reports)

⇨ Already one exceedance would have been a hard contradiction, no exceedance is also only a weak indicator in favor of it.

Solvency backtest – Data from a large German insurance group – 2

Quarterly data from reports in 2016Q1-2022Q3 (see K., Stahl (2023a))

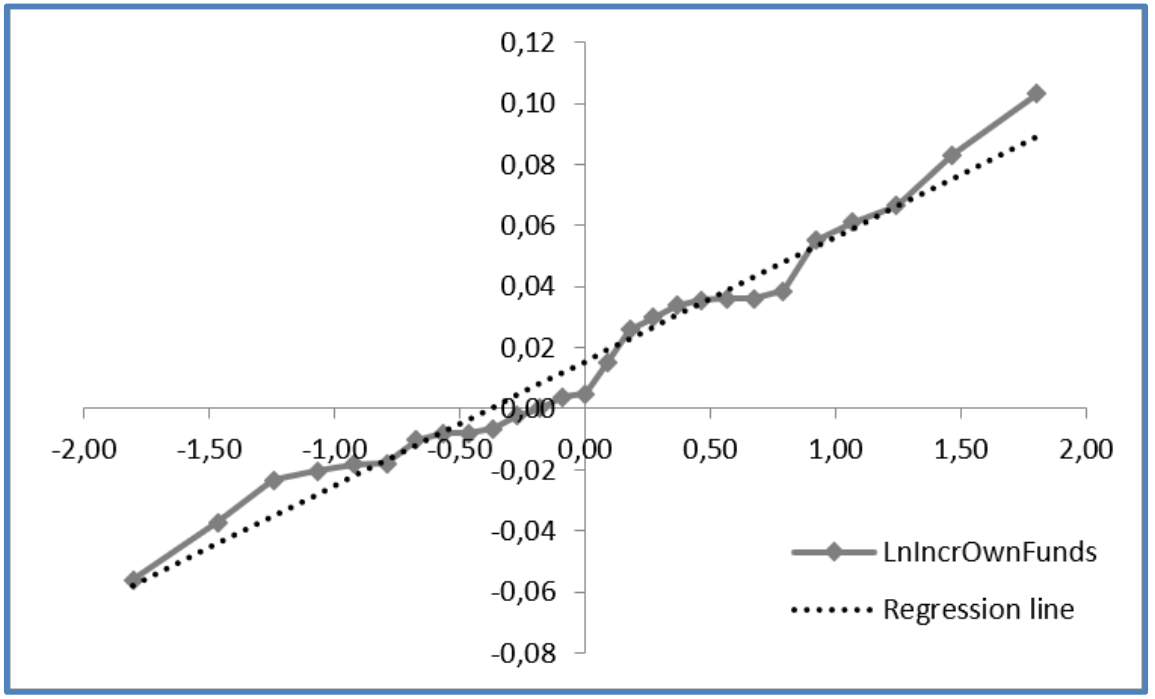


Joint evolution of the **share price** and the **own funds** (correlation ≈ 0.70)

What can you say about the two curves? Which values are based on the better data quality? What was the time of the biggest drop? Where all these data reported in normal times?

Solvency backtest – Data from a large German insurance group – 3

Quarterly data from reports in 2016Q1-2022Q3 (see K., Stahl (2023a))

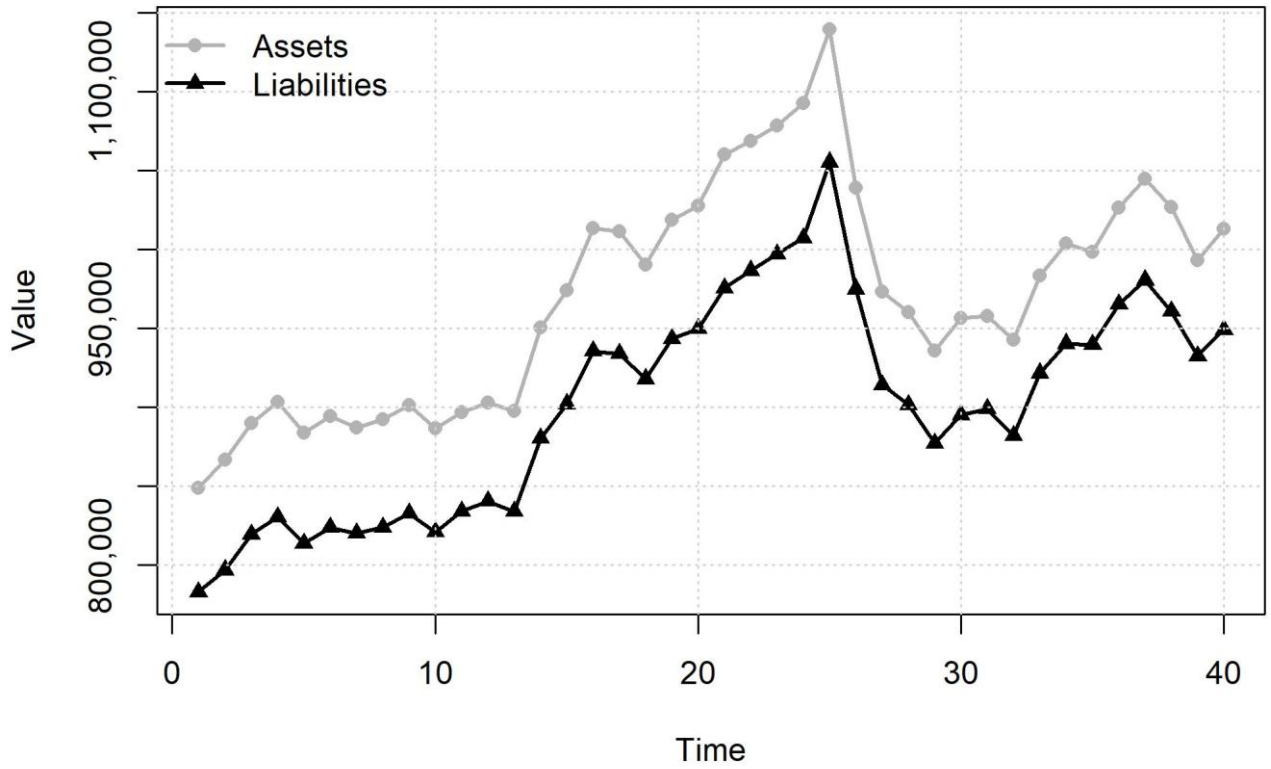


QQ-plot of the increases of log own funds against the N(0,1)-distribution (R^2=0.971)

Comments? Arguments for assuming the own funds to be log-normally distributed? Arguments against it?

Solvency backtest – Data from a large German insurance group – 4

Quarterly data from reports in 2015Q4-2024Q4 (see K. et al. (2026))

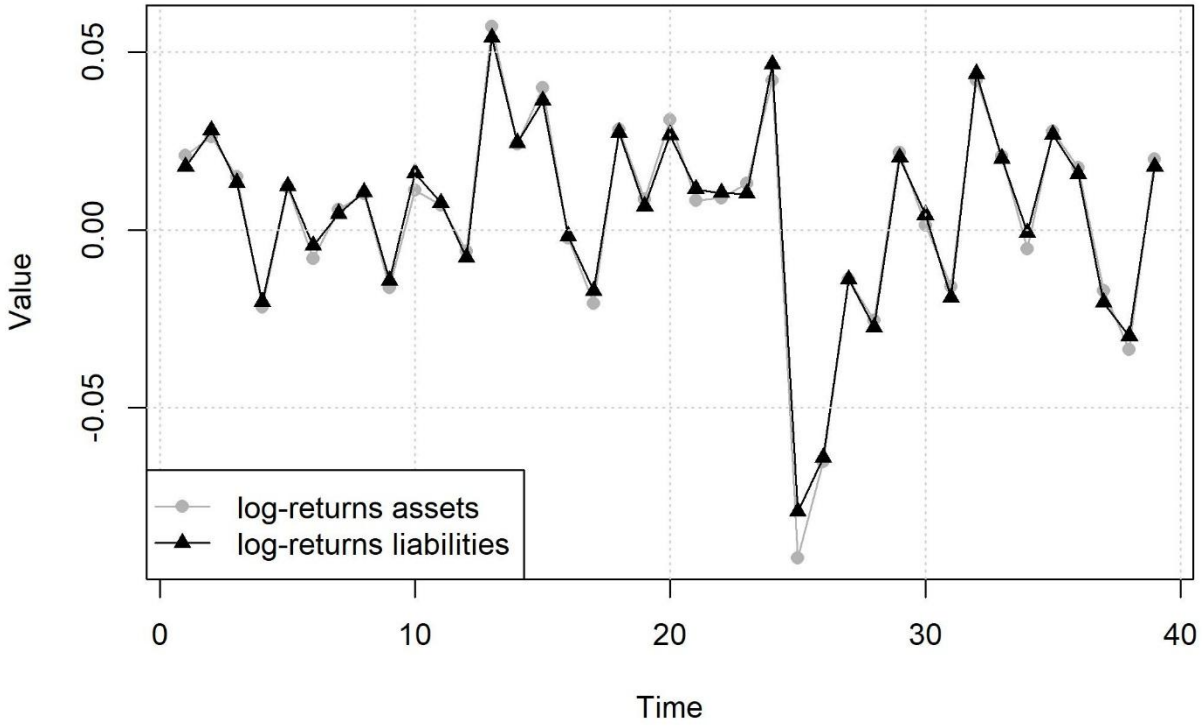


Joint evolution of the **Asset** and the **Liability values** (correlation = 0.998 !!!)

Is this a random effect? Or could this indicate the quality of the management?

Solvency backtest – Data from a large German insurance group – 5

Quarterly data from reports in 2015Q4-2024Q4 (see K. et al. (2026))



Joint evolution of the log **Asset** and **Liability values**

Is this a random effect? Or could this indicate the quality of the management? It might even support the indication off log-normal own funds …

Solvency backtest – Data from a large German insurance group – 6

What have we learnt:

- There are surprisingly many arguments for a log-normal assumption on the own fund dynamics (see the 8 stylised facts in K., Oppold, Schelling, Stahl (2026))
- The extremely close matching of assets and liabilities hints at a high quality of the management with regard to dynamic steering
- Note that the period for backtesting covered the negative interest rate period, the Covid time, the Brexit, the start of the attack on the Ukraine …
- Could we suggest/try a lognormal alternative as a kind of Top-down approach?

$$OF(t+h) = OF(t)e^{\mu h + \sigma W(t)}, \text{ h} > 0$$

- The numbers for the large German insurance group are:

Reported SCR 2021Q3	Calculated SCR on log-normal estimations
10 095 Million Euro	3 238 Million Euro

More discussions on this can be found in K., Stahl (2023b), K. et al. (2026).

Solvency II – Possible dynamic concepts of the SCR

Main concept: Get the SCR from a solution of a portfolio problem

Idea 1: Maximize the lower quantile of the loss over all constant portfolios
Note: The final wealth corresponding to a constant portfolio process π is given by

$$X^{\pi}(T) = xe^{\left(r+\pi(b-r)-\frac{1}{2}\pi^2\sigma^2\right)T+\pi\sigma W(T)}$$

leading to the 0,005 quantile of

$$VaR(X^{\pi}(T)) = xe^{\left(r+\pi(b-r)-\frac{1}{2}\pi^2\sigma^2\right)T+\pi\sigma\sqrt{T}z(0{,}005)}$$

with z(α) being the α-quantile of N(0,1)

- ⇨ As we have z(α)=-2,57 the optimal π is negative for typical values of r, b, σ and small values of T
- ⇨ This does not help us as a negative value of π will typically lead to a decreasing expectation in future wealth and requiring a short position in the risk asset !

Solvency II – Possible dynamic concepts of the SCR - 2

Idea 2: The approach by Basak and Shapiro (2001) "Value-at-Risk Management: Optimal Policies and Asset Prices"

Remember our setting and notation of the martingale method:

$$dB(t) = B(t)rdt, \quad dS_i(t) = S_i(t)\big(b_i dt + \sigma dW(t)\big) \qquad \text{"asset prices"}$$

Our (simplified) portfolio problem:

(25) $$\max_{\pi \in A\prime(x)} E^{0,x}\left(U_2\big(X(T)\big)\right)$$

Needed notation: $I_2(y)$ the inverse of $U'_2(x)$ and further

(27) $$H(t) = exp\left(-\left(r + \frac{1}{2}||\theta||^2\right)t - \theta' W(t)\right)$$

(28) $$\xi(y) = E\left(H(T)I_2\big(yH(T)\big)\right)$$

(29) $$\Upsilon(x) = \xi^{-1}(x)$$

Solvency II – Possible dynamic concepts of the SCR - 3

Remember:

Theorem (Optimal final wealth)

Let x > 0 and let $\xi(y) < \infty$ for all y > 0. Then for

(30) $B^* = I_2\big(\Upsilon(x)H(T)\big)$ "optimal final wealth"

there is a portfolio process $\pi^*(t)$ such that for $\pi^*(t)\epsilon A(x)$ we have

(32) $X^{\pi^*}(T) = B^*$

The setting of Basak and Shapiro:

For a given value $\underline{X}$ ("the floor") we consider an upper bound on the VaR of the form

(BS6) $VaR(\alpha) \leq x - \underline{X}$

that – with the definition of VaR – can be transformed into the "VaR-constraint"

(BS7) $P(X^{\pi}(T) \geq \underline{X}) \geq 1-\propto$

Solvency II – Possible dynamic concepts of the SCR - 3

The Basak and Shapiro problem

(BS8) $$\max_{\pi \in A'(x)} E^{0,x}\left(U_2\left(X^{\pi}(T)\right)\right) \quad \text{s.t.} \quad P(X^{\pi}(T) \geq \underline{X}) \geq 1-\propto$$

Proposition 1 (Basak, Shapiro (2001))

The optimal final wealth of problem (BS8) is given by

(BS9) $$X^{VaR}(T) = \begin{cases} I_2(yH(T)) & \text{if } H(T) < \underline{H} \\ \underline{X} & \text{if } \underline{H} \leq H(T) < \overline{H} \\ I_2(yH(T)) & \text{if } H(T) \geq \overline{H} \end{cases}$$

with y, $\underline{H}, \overline{H}$ being defined as the solutions of

(Con) $\underline{H} := \frac{U_2'(\underline{X})}{y}$, $P\left(H(T) > \overline{H}\right) = \alpha$, and $y \geq 0$ solves $E\left(H(T)X^{VaR}(T;y)\right) = x$.

Further, the VaR-constraint in (BS7) is binding if and only if $\underline{H} < \overline{H}$.

Solvency II – Possible dynamic concepts of the SCR - 4

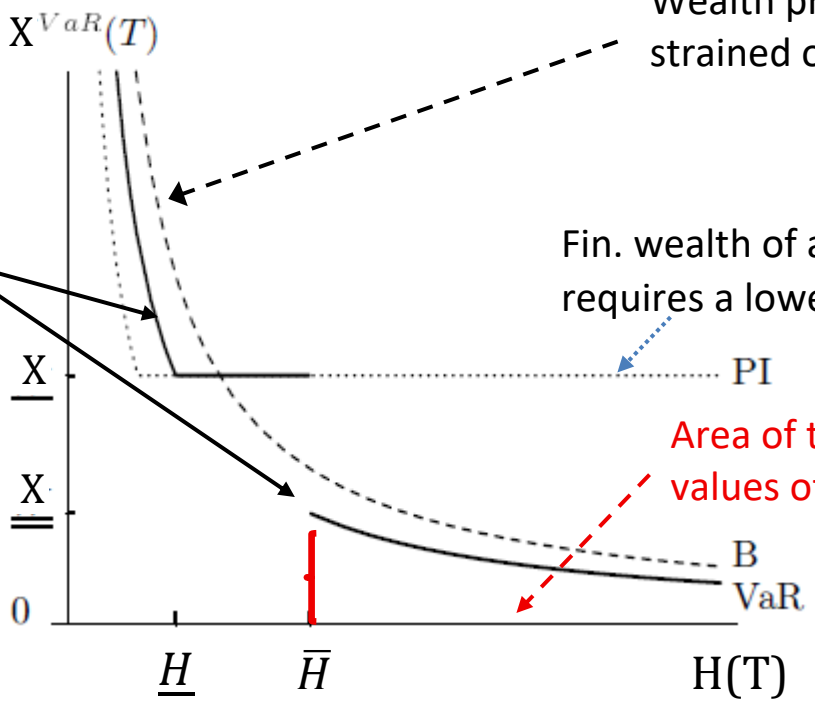
Interpretation of The Basak and Shapiro solution

Wealth process of the uncon-strained optimal strategy

Wealth process of the opti-mal constrained strategy

Fin. wealth of a PI with lower level $\underline{X}$ which requires a lower end value in the good area

$\underline{H}$ is determined as the highest level of H(T) such that the opt. constr. wealth is at its lowest value below $\overline{H}$



Area of the α-quantile of the values of unc. optimal strategy

with $\underline{\underline{X}} := \begin{cases} I_2(y\,\overline{H}) \text{ if } \underline{H} < \overline{H} \\ \underline{H} \text{ otherwise} \end{cases}$

This is a **very dangerous** strategy. Why?

Solvency II – Possible dynamic concepts of the SCR - 5

Final remarks to the Basak and Shapiro solution

a) Basak and Shapiro (2001), Proposition 3, contains explicit formulas for the trading strategies that corresponds to the optimal constraint strategy. It is based on interpreting the final wealth as being made up of the sum of

   a. an unconstrained optimal wealth with a lower initial capital and

   b. a "corridor option" with final payment $\underline{X}$ when we have $H(T) \in \left[\underline{H}, \overline{H}\right]$

b) Basak and Shapiro also report that the choice of an expected shortfall risk measure would lead to a somewhat smoother strategy in the dangerous area than the one corresponding to $X^{VaR}(T)$.

c) The work by Basak and Shapiro (2001) was a landmark paper and opened up the area of research in this direction .

A Final Comment on the Administrational Effort and the Coexistence of Solvency II and IFRS

Stahl (2026) (translated from German): While the first applications of banks for using an internal model in 1997 needed only 6 (!!!) pages, nowadays the solvency report of a large German insurance group consists of 4000 (!!!) pages …

Natural question: Given the presence of the IFRS balance sheet, does a possible information gain justify the work for the Solvency II balance sheet?

A partial answer: In K. et al. (2026), we state:

Stylised Fact 8 (Value of Information) The gain in information (… in addition to the IFRS balance sheet …) associated with the introduction of the solvency balance sheet is small.

This has to be seen in pariticular in the face of high correlations between the two.

Basic literature for this presentation

Basak S., Shapiro A. (2001) Value-at-Risk Management: Optimal Policies and Asset Prices. The Review of Financial Studies, 14 (2), 371–405.

Korn R. and G. Stahl (2023a) A first look back: model performance under Solvency II. European Actuarial Journal, 1–9.

Korn R. and G. Stahl (2023b) Insights from backtesting under Solvency II – Overdose or implicit change of paradigm? Available at SSRN 4643633.

Korn R., Oppold D., Schelling S. and G. Stahl (2026) From Bottom-up or Top-down – How to Manage Dynamics under Solvency II, submitted.

G. Stahl (2026) Solvency II in the New Normal: David gegen Goliath oder Resilienz gegen Autopoiesis. Unpublished Manuscript.